

M003 Extension 2 — English Worksheet (Beginner)

Topic: 3D Mob Heads (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after Week 3) ·

Time: about 30 minutes

Build a mob head in 3D. Now every block needs **three** numbers: **(x, y, z)**. x is **across**, y is **up**, z is **forward** (depth). The heads stand in front of you in the world — look at one as you build.

● Stand on your **home spot** (red block) every run. Move, and the head moves too.

1 · Predict 🧠

Read the steps. Circle or write your answer.

```
place green block at (3, 5, 1)
place green block at (3, 5, 2)
place green block at (3, 5, 3)
```

Only z changes. Circle one: across · up · deeper


```
place green block at (4, 5, 1)
place green block at (4, 6, 1)
```

Only y changes. Circle one: across · up · deeper

2 · Find the Difference 🐛

Correct first, then **broken**. Write one sentence about the bug.

Pair A — This block should be one step deep: (3, 5, 2).

```
# correct  
place green block at (3, 5, 2)
```

```
# broken  
place green block at (3, 5, 1)
```

What is wrong? Which number is different?

Pair B — The eye should be at (4, 7, 2).

```
# correct  
place black block at (4, 7, 2)
```

```
# broken  
place black block at (4, 2, 7)
```

What is wrong? Which two numbers got swapped?

3 · Fill the Gap

To go **deeper**, the **third** number (z) grows. Fill the blanks to go deeper at (3, 5).

```
place green block at (3, 5, 1)
place green block at (3, 5, __)
place green block at (3, 5, __)
```

Word bank: 2 · 3

Write the two numbers:

4 · Build a Head

Look at the  creeper head in the world. Walk around it.

Build the front face first (keep $z = 1$). Read each block's **x** (across) and **y** (up).

Then add depth — build the same face again at $z = 2$, then $z = 3$.

 Tip: the eyes and mouth only show on the **front**. The sides and back are plain.

Send a photo. Then finish these — 3 to 5 sentences.

I built the ...

The lowest front block is at ($x = \dots$, $y = \dots$, $z = \dots$).

To go up I changed the ... number; to go deeper I changed the ... number.

My head is ... blocks deep.

Next time I would ...

Submit 

Send this worksheet + a photo on KakaoTalk.